

Question
cube I , rounds r ,
output bit j

SAT model
does a monomial
trail exist?

no trail

Certificate
bit j is balanced
for *all* round keys

checked: DRAT proof
verified by `drat-trim`

trail found

No certificate
the model cannot
decide the bit

checked: the trail is
re-derived without the
SAT encoding

maximal key pattern,
odd number of trails

Presence proof
bit j is *not* balanced
(nonzero for some keys)

checked: trail validated;
counting cross-checked
with the ANF